LESSON 6.7

Find Factors and Multiples

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the greatest common factor and the least common multiple of two numbers, and use each to solve a real problem.

WHY THIS MATTERS The station quartermaster has two jobs today. First, split the pencil and notebook stock into identical combo packs with nothing left over — that is a factor question. Second, work out when two supply runs that repeat on different cycles will land in the same week — that is a multiple question. Same lesson, opposite tools. Today the stock is 24 pencils and 36 notebooks. One supply run arrives every 6 days and another arrives every 8 days.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Factor *Spanish: Factor*	A whole number that divides another number evenly, with no remainder.	1, 2, 4, 8, 16 divide 16.
Multiple *Spanish: Múltiplo*	The result of multiplying a number by 1, 2, 3, and so on — where you land when you skip-count by it.	Multiples of 8: 8, 16, 24, 32, 40 …
Greatest common factor (GCF) *Spanish: Máximo común divisor (MCD)*	The largest number that is a factor of two or more numbers. It answers “how many equal groups can I make?”	48 and 36 share 1, 2, 3, 4, 6, 12 → GCF = 12
Least common multiple (LCM) *Spanish: Mínimo común múltiplo (mcm)*	The smallest number that is a multiple of two or more numbers. It answers “when will these line up again?”	6, 12, 18, 24 and 8, 16, 24 → LCM = 24
Factor pair *Spanish: Par de factores*	Two factors that multiply to make a number. In a word problem, one is the number of groups and the other is the size of each group.	2 × 12 = 24
Prime factorization *Spanish: Descomposición en factores primos*	Writing a number as a product of prime numbers, usually with a factor tree. Multiplying the primes two numbers share also gives the GCF.	24 = 2 × 2 × 2 × 3

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find the GCF of 20 and 30.

1. Factors of 20: 1, 2, 4, 5, 10, 20.
2. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30.
3. Shared factors: 1, 2, 5, 10. The greatest is 10.

Answer: GCF = 10

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A factor tree breaks each number into primes. 48 = 2 × 2 × 2 × 2 × 3, and 36 = 2 × 2 × 3 × 3.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The factors of _____ are _____ .
- The factors that both numbers share are _____ .
- The greatest common factor is _____ .
- I used the GCF here because the problem asks for _____ .

Word bank factor | factor pair | common factor | greatest common factor | GCF | prime | prime factorization | divides evenly | list | share | greatest | equal groups

WATCH OUT!

A common mistake

A common mistake is reaching for the wrong tool: listing FACTORS for a problem about repeating cycles. Ayasha's classes repeat every 6 and 8 weeks, so the weeks they land on are multiples

LESSON 6.7

Find Factors and Multiples

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the greatest common factor and the least common multiple of two numbers, and use each to solve a real problem.

WHY THIS MATTERS The station quartermaster has two jobs today. First, split the pencil and notebook stock into identical combo packs with nothing left over — that is a factor question. Second, work out when two supply runs that repeat on different cycles will land in the same week — that is a multiple question. Same lesson, opposite tools. Today the stock is 24 pencils and 36 notebooks. One supply run arrives every 6 days and another arrives every 8 days.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find the GCF of 20 and 30.

1. Factors of 20: 1, 2, 4, 5, 10, 20.
2. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30.
3. Shared factors: 1, 2, 5, 10. The greatest is 10.

Answer: GCF = 10

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each number by its relationship to 10.

Groups: A factor of 10 only | A multiple of 10 only | Both a factor and a multiple of 10

_____ 2

_____ 20

_____ 5

_____ 30

_____ 10

_____ 50

STEPS TO TRY

1. Check both relationships: does this number divide 10 evenly, and does 10 divide this number evenly?
2. A positive factor of 10 divides 10 with no remainder.
3. 10 passes both checks, so it belongs in the Both category.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which list shows all the factors of 12?

(A) 1, 2, 3, 4, 6, 12

(B) 1, 12

(C) 2, 3, 4, 6

(D) 12, 24, 36, 48

STEPS TO TRY

1. Ask: what divides 12 with nothing left over?
2. Try each number from 1 up to 12.
3. Do not forget 1 and the number itself.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which list shows the first four multiples of 8?

(A) 1, 2, 4, 8

(B) 2, 4, 6, 8

(C) 8, 16, 24, 32

(D) 8, 9, 10, 11

STEPS TO TRY

1. Skip-count by 8 out loud.
2. 8 × 1, 8 × 2, 8 × 3, 8 × 4.
3. Multiples get bigger; factors stay at or below the number.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the greatest common factor of 12 and 18?

(A) 2
(B) 3
(C) 6
(D) 36

STEPS TO TRY

1. List the factors of each number.
2. Circle the numbers that appear in both lists.
3. Pick the largest circled number.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both problems say "common." How do you decide whether a problem wants the GCF or the LCM?

SENTENCE STARTER If the problem ___ something into equal groups, I use the ___. If it asks when things ___ again, I use the ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help